Pure Discount Instrument: What it Means, How it Works, Example

investopedia.com/terms/p/pure_discount_instrument.asp

What Is a Pure Discount Instrument?

A pure discount instrument is a type of security that pays no income until maturity. Upon expiration, the holder receives the face value of the instrument. The instrument is originally sold for less than its face value—at a discount—and redeemed at par.

Key Takeaways

- Pure discount instruments pay no coupon, instead, paying face value at maturity.
- These instruments are priced at a discount to par. The yield on pure discount instruments, also known as the spot interest rate, is the annualized return that results when the bonds are converted to face value.
- Popular pure discount instruments are zero-coupon bonds and Treasury bills.

How a Pure Discount Instrument Works

Some debt instruments require the issuer to repay the lender the amount borrowed plus interest. This entails making periodic interest payments to the lender until the security matures, at which point the lender is repaid the face value of the security. In other cases, the securities do not make scheduled interest payments. Instead, investors can purchase the securities at a value less than par and receive the face value at maturity. These securities are referred to as pure discount instruments.

Pure discount instruments can take the form of zero-coupon bonds or Treasury bills. The discount on these securities, that is, the difference between the purchase price and the redemption value at maturity, represents the interest that accumulates on these debt instruments. If a pure discount instrument is held to maturity, the bondholder will earn a dollar return equal to the discount.

The difference between the purchase price and par value on pure discount instruments represents the interest an investor earns.

Example of a Pure Discount Instrument

For example, assume a Treasury bill with a face value of $1,000 has time to maturity of 270 days and is currently selling for $950. If the investor holds the T-bill until it matures, they will earn a positive yield of:

r = (Discount / Face value) x (360 / t)

Where,

- r = annualized yield
- Discount = Face value – Purchase price
- 360 = bank convention on the number of days per year
- t = time to maturity

Following our example above, the yield can be calculated as follows:

r = ($50/$1,000) x (360 / 270)

= 0.05 x 1.33

= 0.0665, or 6.65%.

The formula used above is referred to as the bank discount yield. The yield on pure discount instruments is the annualized return that results when the bonds are converted to face value. This yield is also referred to as the spot interest rate. An interest-bearing bond with predictable cash flows or interest payments can be seen as a portfolio of pure discount bonds.

Coupon-bearing bonds are priced using spot rates by assigning the yield of a pure discount instrument maturing in six months to the coupon payment six months from now, the yield of a one-year pure discount instrument to the coupon payments one year from now, and so on, until yields have been assigned to all the bond's cash flows.

The formula for this calculation is as follows:

Price = C1/(1+r1) + C2/(1+r2)2 + C3/(1+r3)3 + … + Cn/(1+rn)n + F/(1+rn)n

Where,

- C = the cash flow for period n
- r = spot rate of interest for period n
- F = face value at maturity

As long as pure discount instruments are available at all maturity terms, the spot rates will accurately reflect the term structure of interest rates.